A student performed simple and vacuum distillations of various mixtures containing two or more of the following compounds: succinic acid ($HOOC(CH_2)_2COOH$), ethyl acetate ($CH_3C(O)OCH_2CH_3$), *n*-butanol ($CH_3(CH_2)_3OH$), butyric acid ($CH_3(CH_2)_2COOH$), butanone ($CH_3C(O)CH_2CH_3$), butyraldehyde ($HC(O)(CH_2)_2CH_3$), *tert*-butanol ($(CH_3)_3COH$), and diethyl ether ($CH_3CH_2OCH_2CH_3$). The boiling points of some of these compounds are listed in Table 1.

Table 1 Boiling Points of Selected Organic Compounds

Compound	Boiling point (°C)
$CH_3CH_2OCH_2CH_3$	34.6
$HC(O)(CH_2)_2CH_3$	74.8
$CH_3C(O)CH_2CH_3$	79.6
$(CH_3)_3COH$	83.0
$CH_3(CH_2)_2COOH$	163
$HOOC(CH_2)_2COOH$	235

The student used gas-liquid chromatography (gc) to analyze a mixture containing three of the compounds listed above. Figure 1 shows the resulting gc trace.

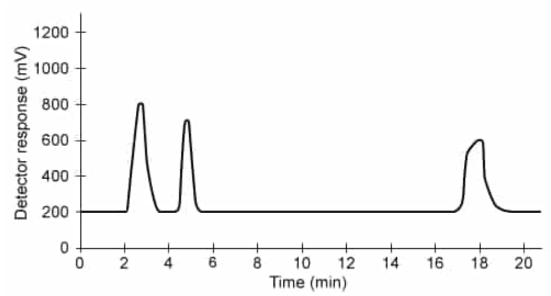


Figure 1 The gc trace from the separation of a mixture containing three organic compounds

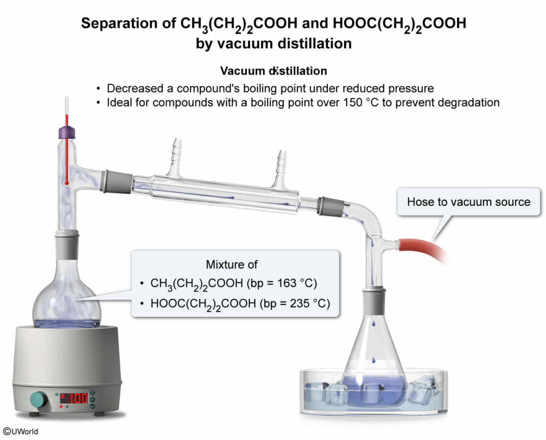
The student separates a mixture of $CH_3(CH_2)_2COOH$ and $HOOC(CH_2)_2COOH$ by vacuum distillation because:

I. compounds with a boiling point over 150 °C could degrade at their boiling point.
II. vacuum distillation requires compounds with boiling points less than 25 °C apart.
III. distillation under a vacuum increases the boiling point of a compound.
IV. distillation under a vacuum decreases the boiling point of a compound.

- A. I and II only [3%]
- B. I and III only [6%]
- ✔ C. I and IV only [81%]
- D. II and IV only [8%]

Correct
82% answered correctly

Explanation:



Distillation is a technique used to separate compounds based on their boiling point (ie, the temperature in which the vapor pressure equals the pressure of the system). There are three common types of distillation: simple, fractional, and vacuum. The type of distillation appropriate for a mixture depends on the boiling points of the mixture components.

Vacuum distillation is performed under reduced system pressure, thereby **lowering** a compound's **boiling point** (bp) relative to its boiling point at atmospheric pressure **(Number IV)**. Compounds with high boiling points (>150 °C) tend to decompose at or near their boiling point. Therefore, vacuum distillation is ideal for compounds with a **boiling point over 150 °C** to **prevent degradation (Number I)**.

Table 1 indicates that the boiling points of $CH_3(CH_2)_2COOH$ and $HOOC(CH_2)_2COOH$ are 163 °C and 235 °C, respectively. Because these compounds have boiling points over 150 °C, vacuum distillation is a suitable separation technique because the vacuum lowers the boiling points of $CH_3(CH_2)_2COOH$ and $HOOC(CH_2)_2COOH$, preventing degradation during the distillation.

(Number II) *Fractional* distillation is ideal for compounds with boiling points less than 25 °C apart. Based on the information in Table 1, the boiling points of $CH_3(CH_2)_2COOH$ and $HOOC(CH_2)_2COOH$ are *greater* than 25 °C apart.

(Number III) Because vacuum distillation is done at a reduced pressure (ie, lower than atmospheric pressure), a compound's boiling point *decreases* relative to atmospheric pressure.

Educational objective:
Vacuum distillation is performed under reduced pressure, lowering a compound's boiling point relative to its boiling point at atmospheric pressure. Vacuum distillation is ideal for compounds with a boiling point above 150 °C to prevent degradation.

Time spent:0 sec	**QID**:403278
Subject:	**System:**
Organic Chemistry	Separation Techniques, Spectroscopy, and Analytical Methods

A student performed simple and vacuum distillations of various mixtures containing two or more of the following compounds: succinic acid ($HOOC(CH_2)_2COOH$), ethyl acetate ($CH_3C(O)OCH_2CH_3$), *n*-butanol ($CH_3(CH_2)_3OH$), butyric acid ($CH_3(CH_2)_2COOH$), butanone ($CH_3C(O)CH_2CH_3$), butyraldehyde ($HC(O)(CH_2)_2CH_3$), *tert*-butanol ($(CH_3)_3COH$), and diethyl ether ($CH_3CH_2OCH_2CH_3$). The boiling points of some of these compounds are listed in Table 1.

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The student used gas-liquid chromatography (gc) to analyze a mixture containing three of the compounds listed above. Figure 1 shows the resulting gc trace.

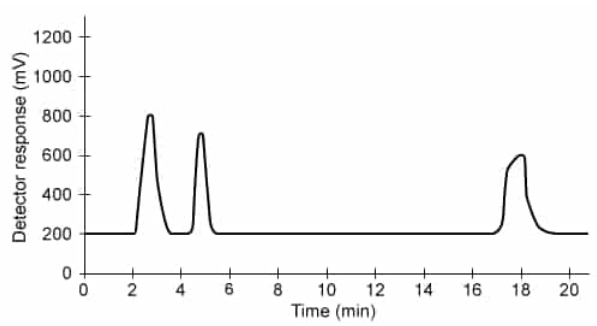


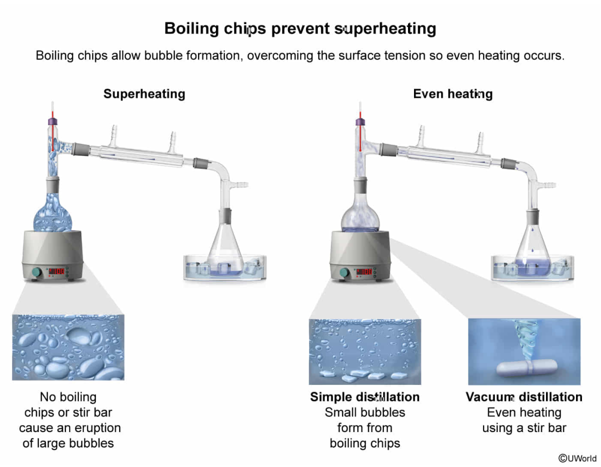
Figure 1 The gc trace from the separation of a mixture containing three organic compounds

The student uses boiling chips in the simple distillation and a stir bar in the vacuum distillation. Why are the boiling chips and stir bar necessary in the distillations?

- ✔ ◉ A. To provide sites for bubble formation, overcoming the surface tension so the liquid boils evenly [70%]
- ○ B. To increase the vapor pressure, causing superheating to occur [4%]
- ○ C. To introduce nucleation sites, causing large bubbles to erupt [12%]
- ○ D. To decrease the distillation rate, resulting in better separation of the mixture components [12%]

Correct
70% answered correctly

Explanation:



Superheating occurs when a liquid is heated past its boiling point but does not boil. Surface tension can prevent the formation of bubbles and cause superheating to occur. When bubbles try to form, the surface tension can cause an increase in the local vapor pressure that goes beyond the ambient pressure, allowing the liquid to heat past its boiling point. This can cause large bubbles to form suddenly and erupt from the distillation flask (ie, an effect known as bumping).

Boiling chips, which are made of a porous material, provide surfaces with nucleation sites where small bubbles can form. This overcomes the surface tension and allows the liquid to **boil evenly**, preventing superheating. Boiling chips are used in simple and fractional distillations but cannot be used in vacuum distillations because the vacuum removes the air from the boiling chips. Therefore, magnetic **stir bars** are used in place of boiling chips in vacuum distillations to ensure even heating and **prevent superheating**.

Therefore, the student used boiling chips in the simple distillation and a stir bar in the vacuum distillation to provide sites for bubble formation, which overcomes the surface tension and allows the liquid to boil evenly.

(Choice B) Boiling chips and stir bars are added to *prevent* superheating and do not cause vapor pressure to increase.

(Choice C) Boiling chips introduce nucleation sites, but this causes *small* bubbles to form. Large bubbles erupting from the distillation flask occurs when boiling chips are not used.

(Choice D) Boiling chips and stir bars do not affect the rate of distillation.

Educational objective:
Boiling chips are porous materials used in simple and fractional distillations that introduce nucleation sites where small bubbles form, overcoming the surface tension

and allowing even boiling (ie, prevents superheating). Stir bars are used in place of boiling chips in vacuum distillations to prevent superheating.

Time spent: 0 sec

Subject: Organic Chemistry

QID:403279

System: Separation Techniques, Spectroscopy, and Analytical Methods

A student performed simple and vacuum distillations of various mixtures containing two or more of the following compounds: succinic acid ($HOOC(CH_2)_2COOH$), ethyl acetate ($CH_3C(O)OCH_2CH_3$), *n*-butanol ($CH_3(CH_2)_3OH$), butyric acid ($CH_3(CH_2)_2COOH$), butanone ($CH_3C(O)CH_2CH_3$), butyraldehyde ($HC(O)(CH_2)_2CH_3$), *tert*-butanol ($(CH_3)_3COH$), and diethyl ether ($CH_3CH_2OCH_2CH_3$). The boiling points of some of these compounds are listed in Table 1.

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$(CH_3)_3COH$	83.0
$CH_3(CH_2)_2COOH$	163
$HOOC(CH_2)_2COOH$	235

The student used gas-liquid chromatography (gc) to analyze a mixture containing three of the compounds listed above. Figure 1 shows the resulting gc trace.

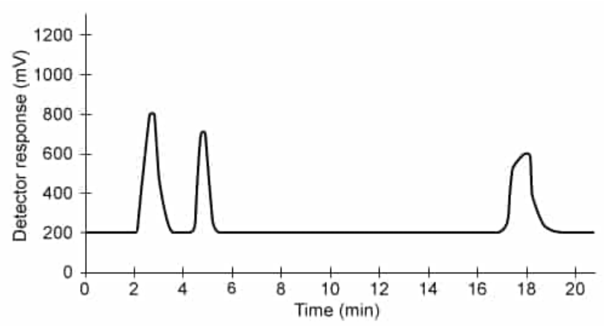


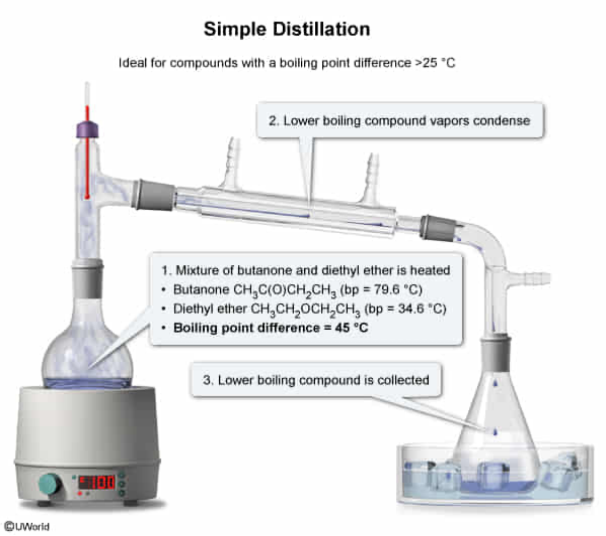
Figure 1 The gc trace from the separation of a mixture containing three organic compounds

Based on the information in Table 1, what pair of compounds can be separated by simple distillation?

- ◯ A. Butyraldehyde and butanone [9%]
- ◯ B. *t*-butanol and butyraldehyde [6%]
- ◯ C. Butanone and *t*-butanol [7%]
- ✔ ◉ D. Butanone and diethyl ether [76%]

Correct
77% answered correctly

Explanation:



Distillation is a technique used to separate liquid mixtures based on the boiling points of the mixture components. The compounds distill in order of increasing boiling point (ie, the lowest boiling point will evaporate first). The mixture is heated slowly so that the boiling temperatures of the higher boiling compounds are not reached before the condensed vapors (ie, the distillate) of the lower boiling compound is collected. The vapors of the compound travel up the distillation column to the thermometer and condenser and then condense into the receiving flask, which is in an ice bath to assist in condensation and keep the distillate from evaporating.

There are three common types of distillation: simple, fractional, and vacuum. The appropriate type of distillation for a mixture depends on the boiling points of the mixture components:

- **Simple distillation** (boiling points **<150 °C** and **>25 °C apart**)
- Fractional distillation (boiling points <150 °C and <25 °C apart)
- Vacuum distillation (boiling points >150 °C)

Based on the information in Table 1, butanone, $CH_3C(O)CH_2CH_3$ (boiling point = 79.6 °C), and diethyl ether, $CH_3CH_2OCH_2CH_3$ (boiling point = 34.6 °C), are the only pair of compounds given in the question with boiling points greater than 25 °C apart (boiling point difference = 79.6 °C − 34.6 °C = 45 °C). Because simple distillations are ideal for compounds with boiling points greater than 25 °C apart, butanone and diethyl ether can be separated by simple distillation.

(Choices A, B, and C) All these pairs of compounds have a boiling point difference *less* than 25 °C apart; therefore, simple distillation is *not* a suitable technique to separate these pairs of compounds. Fractional distillation is ideal for mixtures in which the components have boiling point differences less than 25 °C apart.

Educational objective:
Distillation is a technique that separates liquid mixtures based on the boiling points of the mixture components. The correct type of distillation to use depends on the boiling points of the mixture components: simple (boiling points <150 °C and >25 °C apart), fractional (boiling points <150 °C and <25 °C apart), and vacuum (boiling points >150 °C).

Time spent: 0 sec

Subject:
Organic Chemistry

QID:403280

System:
Separation Techniques, Spectroscopy, and Analytical Methods

A student performed simple and vacuum distillations of various mixtures containing two or more of the following compounds: succinic acid ($HOOC(CH_2)_2COOH$), ethyl acetate ($CH_3C(O)OCH_2CH_3$), *n*-butanol ($CH_3(CH_2)_3OH$), butyric acid ($CH_3(CH_2)_2COOH$), butanone ($CH_3C(O)CH_2CH_3$), butyraldehyde ($HC(O)(CH_2)_2CH_3$), *tert*-butanol ($(CH_3)_3COH$), and diethyl ether ($CH_3CH_2OCH_2CH_3$). The boiling points of some of these compounds are listed in Table 1.

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$HOOC(CH_2)_2COOH$	235

The student used gas-liquid chromatography (gc) to analyze a mixture containing three of the compounds listed above. Figure 1 shows the resulting gc trace.

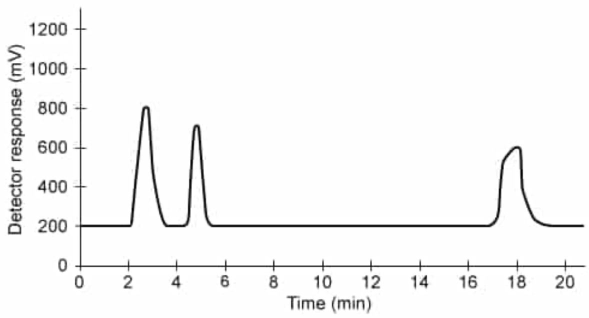


Figure 1 The gc trace from the separation of a mixture containing three organic compounds

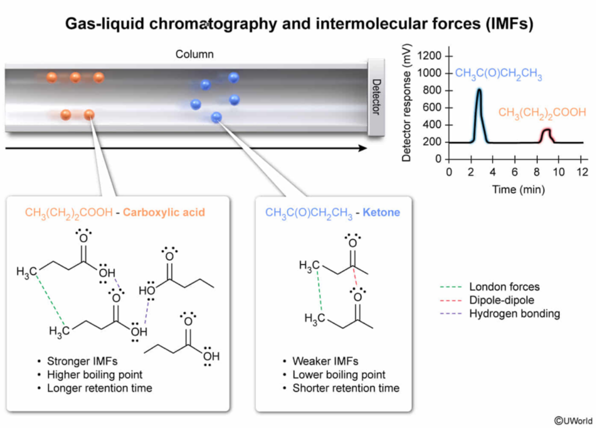
If a gc trace of butanone shows a contaminant with a much longer retention time, which of the following compounds might be the contaminant?

- ○ A. $HC(O)(CH_2)_2CH_3$ [6%]
- ✔ ◉ B. $CH_3(CH_2)_2COOH$ [77%]
- ○ C. $CH_3CH_2OCH_2CH_3$ [9%]
- ○ D. $CH_3C(O)OCH_2CH_3$ [7%]

Correct
76% answered correctly

76% answered correctly

Explanation:



Gas-liquid chromatography is a technique used to separate compounds based on boiling point, which depends on several factors, including molecular weight and intermolecular forces. A small amount of a liquid mixture is injected into the gas chromatograph, and the compounds are vaporized by heating. The most volatile (lowest boiling) compound will vaporize rapidly, move through the column quickly, and have the shortest retention time. Higher boiling compounds can condense and interact with the liquid stationary phase longer (ie, move more slowly through the column), resulting in longer retention times.

Based on the information in Table 1, butanone has a boiling point of 79.6 °C and a formula of $CH_3C(O)CH_2CH_3$. This indicates butanone is a ketone and experiences dipole-dipole interactions and weaker London forces. Because the possible contaminants all contain the same number of carbon atoms, the type of functional group and intermolecular forces must be examined.

- $HC(O)(CH_2)_2CH_3$ is an aldehyde.
- $CH_3(CH_2)_2COOH$ is a carboxylic acid.
- $CH_3CH_2OCH_2CH_3$ is an ether.
- $CH_3C(O)OCH_2CH_3$ is an ester.

Of these functional groups, carboxylic acids experience the strongest intermolecular forces (hydrogen bonding) along with weaker dipole-dipole interactions and London forces. As a result, the carboxylic acid should have the highest boiling point, as confirmed by the measured boiling points of $CH_3(CH_2)_2COOH$ (163 °C) and butanone (79.6 °C) in Table 1. Because a higher boiling point corresponds to a longer retention time, $CH_3(CH_2)_2COOH$ is a possible contaminant in the gc trace of butanone.

(Choices A and D) Like butanone, $HC(O)(CH_2)_2CH_3$ and $CH_3C(O)OCH_2CH_3$ have similar boiling points because they experience dipole-dipole interactions and London forces. Therefore, neither compound could have a much longer retention time than butanone.

(Choice C) Table 1 indicates that $CH_3CH_2OCH_2CH_3$ has a *lower* boiling point than butanone (34.6 °C versus 79.6 °C). Therefore, $CH_3CH_2OCH_2CH_3$ has a *shorter* retention time than butanone.

Educational objective:
Gas-liquid chromatography is a technique that separates compounds based on boiling point. Compounds with lower boiling points have shorter retention times than compounds with higher boiling points. For compounds with a similar number of carbon atoms, the boiling point depends on the functional group and the strength of intermolecular forces present.

Time spent:0 sec

Subject:
Organic Chemistry

QID:403281

System:
Separation Techniques, Spectroscopy, and Analytical Methods

A student performed simple and vacuum distillations of various mixtures containing two or more of the following compounds: succinic acid ($HOOC(CH_2)_2COOH$), ethyl acetate ($CH_3C(O)OCH_2CH_3$), *n*-butanol ($CH_3(CH_2)_3OH$), butyric acid ($CH_3(CH_2)_2COOH$), butanone ($CH_3C(O)CH_2CH_3$), butyraldehyde ($HC(O)(CH_2)_2CH_3$), *tert*-butanol ($(CH_3)_3COH$), and diethyl ether ($CH_3CH_2OCH_2CH_3$). The boiling points of some of these compounds are listed in Table 1.

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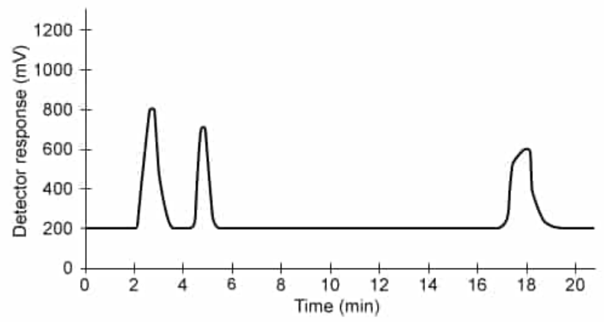


Figure 1 The gc trace from the separation of a mixture containing three organic compounds

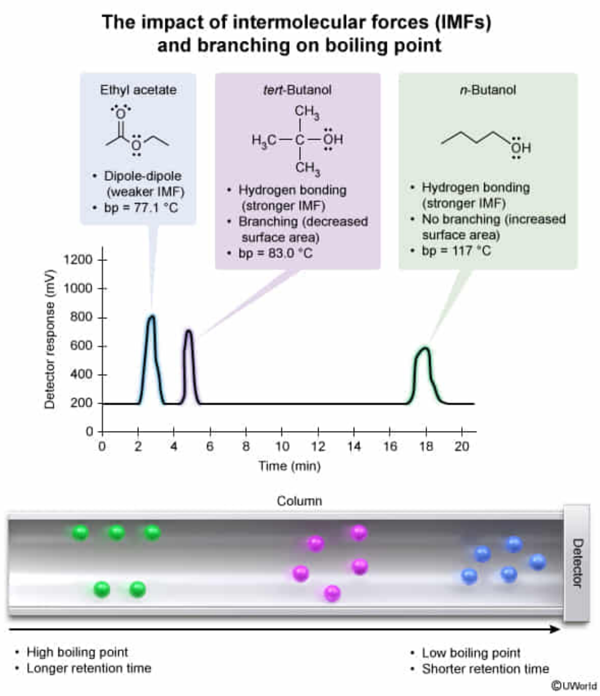
Suppose that the gc trace in Figure 1 resulted from the separation of a mixture of ethyl acetate, *n*-butanol, and *tert*-butanol. What is the expected order of elution of these compounds from first to last?

- ○ A. Ethyl acetate → *n*-butanol → *tert*-butanol [25%]
- ✔ ● B. Ethyl acetate → *tert*-butanol → *n*-butanol [37%]
- ○ C. *n*-Butanol → *tert*-Butanol → ethyl acetate [17%]
- ○ D. *tert*-Butanol → *n*-butanol → ethyl acetate [19%]

Correct

38% answered correctly

Explanation:



Gas-liquid chromatography is a technique used to **separate** compounds based on **boiling point**, which depends on several factors, including molecular weight, intermolecular forces, and branching. The compound with the lowest boiling point vaporizes rapidly and moves through the column quickly. However, higher boiling compounds can condense and interact more with the liquid stationary phase, moving through the column more slowly.

Comparing the structures of ethyl acetate, *n*-butanol, and *tert*-butanol:

- Ethyl acetate ($CH_3C(O)OCH_2CH_3$) is a four-carbon ester.
- *n*-Butanol ($CH_3(CH_2)_3OH$) and *tert*-butanol ($(CH_3)_3COH$) are both four-carbon alcohols.

Compounds with the same number of carbon atoms follow a pattern of increasing intermolecular forces and boiling point based on the functional group in the compound. Esters experience dipole-dipole interactions and London forces, whereas alcohols also experience strong hydrogen bonding and the weaker dipole-dipole interactions and London forces. Because ethyl acetate experiences the weakest intermolecular forces of the given molecules, it will have the shortest retention time and elute first.

n-Butanol and *tert*-butanol have the **same number of carbon atoms** and contain the **same functional group** but differ in the amount of branching. *n*-Butanol does not have any branching and has a larger surface area whereas *tert*-butanol is **highly branched** and has a **smaller surface area**. Less surface area results in fewer

London forces and **decreases the boiling point**. As such, *tert*-butanol has a lower boiling point and elutes before *n*-butanol.

Therefore, the order of elution from first to last is ethyl acetate → *tert*-butanol → *n*-butanol.

(Choice A) *n*-Butanol will elute *after tert*-butanol because it has a *larger* surface area (less branching) and therefore has a *higher* boiling point.

(Choice C) This order of elution is *reversed* (ie, from last to first).

(Choice D) Ethyl acetate experiences the weakest intermolecular forces of the three compounds. Therefore, it has the *lowest* boiling point and will elute *first*.

Educational objective:
Gas-liquid chromatography is a technique that separates compounds based on boiling point. Lower boiling compounds have shorter retention times than higher boiling compounds. For compounds with the same functional group and same number of carbons, branching (smaller surface area) decreases the boiling point.

Time spent:0 sec

Subject:
Organic Chemistry

QID:403282

System:
Separation Techniques, Spectroscopy, and Analytical Methods